Constituency Tests for Concerned Representation in Minimal Agents

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Abstract

Arc 1 of the maintained-concern program showed that a minimal homeostatic agent can detect boundary staleness, allocate costly null probes, saturate after identification, and re-engage after regime shifts. It also found a precise ceiling: under a shared-head, null-only intervention regime, better probe policy no longer closes role-specific mediated identifiability. This paper opens Arc 2A by asking whether the next intervention problem is syntactic: can a concerned agent discover which parts of a world belong together causally?

We introduce the **Concerned Shape Grammar**, a symbolic benchmark inspired by constituency tests from cognitive work on geometric shape syntax. A visible six-part shape can have the same surface but different hidden parse trees. Causal roles such as shield, poison, repair, core, food, and trap interact only when they are bound inside the same constituent. The agent must decide whether to pay for an intervention that reveals the parse, and it must avoid probing low-concern ambiguity that does not affect viability.

In a 200-trial deterministic design pilot, a 5,000-trial symbolic Modal sweep, a learned 5-seed Modal sweep, a vector-observation 5-seed Modal sweep, a pixel-rendered 5-seed local sweep, a pixel-level intervention-invention 5-seed Modal sweep, and a rich intervention-program 5-seed Modal sweep, the concern-gated syntax agent is the only agent family that passes the full gate. The first learned agent receives candidate parses as visible hypotheses but never receives the hidden true parse at test time. The vector-observation agent then removes those candidate-parse features: the visible coordinate surface is invariant under swapping the hidden true and alternate parse. The pixel-rendered agent removes vector parts as a given representation: it receives 48x48 RGB images, extracts objects by connected components, and learns from centroids, colors, areas, and shape density. Across five pixel seeds, the concerned pixel agent reaches high-concern parse accuracy 0.996, action accuracy 0.999, subtree accuracy 0.786, object extraction rate 1.000, low-concern probe rate 0.187, and gate pass rate 1.000. The intervention-invention gate then makes the agent choose among `observe_pair(a,b)` programs rather than receiving the causal target. Across five Modal seeds, the concerned program inventor reaches high-concern parse accuracy 1.000, action accuracy 1.000, target accuracy 1.000, useful program rate 1.000, low-concern probe rate 0.156, and gate pass rate 1.000. A target-only agent identifies the right pair but probes low concern at 1.000; a concern-without-target agent probes at the right times but targets only 0.088 of high-concern cases. The rich-program gate then requires choosing among `observe_pair`, `move_anchor`, `ablate_pair`, and composed move+observe families. Across five Modal seeds, the concerned program composer reaches family accuracy 1.000, target accuracy 1.000, useful-program rate 1.000, rich-program rate 1.000, low-concern program rate 0.162, and gate PASS. Finally, a held-out v2 transfer repair makes role-kind and true-parse transfer explicit. Across five Modal seeds, an explicit role-equivariant rich world model reaches transfer gate 1.000, family/target/useful/rich 1.000, low-concern program rate 0.000, and gate PASS, while the learned rich composer and partial controls fail distinct slices. A supervised learned slot-semantics repair then replaces the explicit RGB role decoder with role-token prototypes and preserves the same transfer gate across five Modal confirmation seeds (600 train, 240 test, 30 epochs per seed), with semantic kind/pair accuracy 1.000 and low-concern program rate 0.000. Shortcut reward, passive perceptual inference, no-tree planning, random program probing, family-only selection, target-only selection, rich programs without concern, and restless inquiry all fail for different anti-cheat reasons. The result is still not a claim about human cognition or learned unsupervised object slots. It is an accepted Phase 2A learned-mechanism surface: **reward is not syntax,**

compression is not syntax, probe availability is not intervention invention, target selection is not program composition, explicit role decoding is not required for supervised slot semantics, and uncertainty reduction is not concerned inquiry.

1. Why Arc 2A Exists

The Metric Stack of Concern ended with a positive mechanism and a ceiling. The positive mechanism was a detect-allocate-saturate-re-engage cycle for self/world attribution in minimal homeostatic agents. The ceiling was equally important: under null-only intervention and shared mediated heads, the agent could predict total world response while failing to identify role-specific mediated components.

That ceiling has two sides. Arc 2B studies the body side: what architectures can express the required distinctions? Arc 2A studies the intervention side: what actions make the world's hidden grammar visible?

The motivation comes from a nearby but distinct empirical result. Revencu, Pajot, and Dehaene (2026) argue that adult geometric-shape representations show syntactic structure. Their key methodological move is not merely to show that shape complexity predicts behavior. They replace compression proxies with constituency tests: structural ambiguity, subtree facilitation, and syntactic movement. They also report that current neural networks can partially solve match/deviant tasks without showing the same syntactic effects.

This paper imports that methodological lesson into the concern program. The claim is not that minimal agents have human visual syntax. The claim is that the Metric Stack needs an analogous gate:

> Performance is not constituency. Compression is not constituency. Uncertainty reduction is not concerned inquiry.

Recent object-centric and neuro-symbolic work also shaped the pixel gate. Causal-JEPA uses object-level latent interventions to force interaction-aware world-model dependencies rather than letting predictors fall back on self-dynamics. Object-centric causal-representation work warns that multi-object observations break simple injective vector assumptions, making entity factorization part of the causal-learning problem. Neuro-symbolic ARC systems similarly separate perception, object abstraction, hypothesis proposal, and consistency filtering. The pixel gate below imports the modest version of that lesson: before claiming syntax from pixels, first require an auditable object extraction layer and show that passive object attributes are still insufficient without concern-gated intervention.

2. Benchmark

Each trial has a six-part visible shape. The visible role sequence is fixed, but two hidden parse trees can organize the same surface differently. A parse is a pair of high-level constituents, each containing three leaves.

Examples of parse candidates:

```
repeat_concat    = (0,1,2) | (3,4,5)
hooked_repeat    = (0,1,3) | (2,4,5)
alternating_bind = (0,2,4) | (1,3,5)
edge_core        = (0,4,5) | (1,2,3)
```

Causal roles interact only when they belong to the same constituent:

Role pair	Causal rule

shield + poison	shield reduces poison damage only within a subtree
repair + core	repair reduces core damage only within a subtree
food + trap	trap contaminates food only within a subtree
signal + ornament	structural ambiguity with no viability consequence

The low-concern case is essential. Without it, a generic uncertainty reducer could look successful simply by probing every ambiguity.

3. Interventions

The intervention language contains five operations:

Intervention	Purpose
`null`	no information, no cost
`pair_probe`	test whether the causal role pair is in the same subtree
`distractor_pair_probe`	matched non-informative pair probe
`high_constituent_move`	test which high-level group moves as a unit
`role_ablation`	observe the viability signature of the causal constituent

The central selector, `concerned_syntax`, chooses an intervention by expected parse information weighted by the viability gap, minus intervention cost. It has a hard no-restless-inquiry threshold: if the parse gap is below 0.10, it chooses `null`.

4. Selectors

Selector	What it tests
`null_policy`	behavior without inquiry
`flat_valence`	roles without constituency
`compression_proxy`	shortest parse without intervention
`uncertainty_only`	information gain without concern
`concerned_syntax`	information gain gated by viability relevance

This is the anti-cheat surface. A selector can be good at action while failing parse. It can recover parse while probing too much. It can choose a compressed parse while missing the causally relevant one.

5. Pilot Result

Local command:

```
python3 -m experiments.concerned_syntax.benchmark \
  --trials 200 --seed 20260616 \
  --out artifacts/concerned_syntax/pilot.json \
  --report experiments/concerned_syntax/results/pilot_2026_06_16.md
```

Summary:

Selector	Parse high	Action	Subtree	High probe	Low probe	Mean regret	Gate
compression_proxy	0.523	0.890	0.550	0.000	0.000	0.055	fail
concerned_syntax	1.000	1.000	0.805	1.000	0.000	0.001	PASS
flat_valence	0.000	0.920	0.540	0.000	0.000	0.074	fail
null_policy	0.523	0.890	0.550	0.000	0.000	0.055	fail
uncertainty_only	1.000	1.000	1.000	1.000	1.000	0.000	fail

The result has the desired diagnostic pattern:

1. `flat_valence` has high action accuracy but fails parse and subtree gates.
2. `compression_proxy` sometimes guesses the right parse, but does not intervene and fails the mechanistic gate.
3. `uncertainty_only` recovers parse perfectly but probes all low-concern ambiguity, failing no-restless-inquiry.
4. `concerned_syntax` probes every high-concern ambiguity, avoids low-concern probing, and preserves action.

6. Discovery-Regime Status

This is a regime transition relative to Arc 1. Arc 1 represented concern as viability prediction, valence, self/world attribution, probe value, and component identifiability. Arc 2A adds a new artifact type: **causal constituency under concern**.

That makes the result discovery-leaning, but still benchmark-level. The pilot does not show that a neural agent has learned syntax. It shows that the repo now has a task surface on which syntax, compression, uncertainty, and concern can dissociate.

7. Modal Multi-Seed Sweep

The multi-seed sweep was run remotely through Modal:

```
doppler --scope /Users/jawaun/superoptimizers run -- \
  uvx --python 3.12 --from modal modal run \
  experiments/concerned_syntax/modal_concerned_syntax_sweep.py \
  --trials 1000
```

The sweep used five seeds and 1,000 shape trials per seed. Raw JSON remains local under `artifacts/concerned_syntax/`; the public report is `experiments/concerned_syntax/results/modal_sweep_2026_06_16.md`.

Summary:

Selector	Parse high	Action	Subtree	High probe	Low probe	Mean regret	Gate pass rate
compression_proxy	0.560	0.891	0.583	0.000	0.000	0.048	0.000
concerned_syntax	1.000	1.000	0.808	1.000	0.000	0.003	1.000
flat_valence	0.000	0.876	0.503	0.000	0.000	0.066	0.000
null_policy	0.560	0.891	0.583	0.000	0.000	0.048	0.000
uncertainty_only	1.000	1.000	1.000	1.000	1.000	0.000	0.000

This replicates the design-pilot pattern across seeds. The key result is not that `concerned_syntax` has the best reward; `uncertainty_only` has zero regret too. The key result is that only concern-weighted syntax passes both the positive inquiry gate and the no-restless-inquiry gate.

8. Learned-Agent Gate

The next gate removes the hand-coded selector. Candidate parses remain visible as hypotheses, as in a structural-ambiguity task, but the true parse is hidden at test time. A small learned agent trains three binary components:

- a concern-gated intervention policy;
- a parse interpreter that binds pair-probe observations to candidate constituents;
- a shortcut action head used as a reward-only control.

The tree-binding body receives candidate-subtree equality features for the probed pair. The no-tree control receives the same surface roles, pair identity, concern weight, and probe observation, but not the candidate-binding features needed to attach that observation to a constituent. The guarded learner also gets a deterministic 20% low-concern calibration budget. This is below the 0.25 anti-restless cap and prevents syntax maintenance from failing exactly at the subtree threshold.

Remote command:

```
doppler --scope /Users/jawaun/superoptimizers run -- \
  uvx --python 3.12 --from modal modal run \
  experiments/concerned_syntax/modal_learned_agents_sweep.py \
  --train-trials 3000 --test-trials 1200 --epochs 90
```

Summary:

Agent	Parse high	Action	Subtree	High probe	Low probe	Gate
guarded syntax	1.000	1.000	0.797	1.000	0.202	PASS
no-tree planner	0.492	0.875	0.494	1.000	0.000	fail
restless tree	1.000	1.000	1.000	1.000	1.000	fail

shortcut reward	0.494	0.880	0.495	0.000	0.000	fail

The full public report also records mean probe cost and regret.

This is the first learned Phase 2A result. It shows that a learner can pass the concerned-syntax gate without hidden parse access, but only when it has both the tree-binding feature needed for causal constituency and the formal guard needed to prevent restless low-concern inquiry.

The next version should add richer learned agents:

- tree-structured model
- flat MLP baseline
- object-slot baseline
- learned intervention policy
- role/parse held-out generalization
- neural anti-cheat probes for parse, subtree, and intervention usefulness

9. Vector-Observation Gate

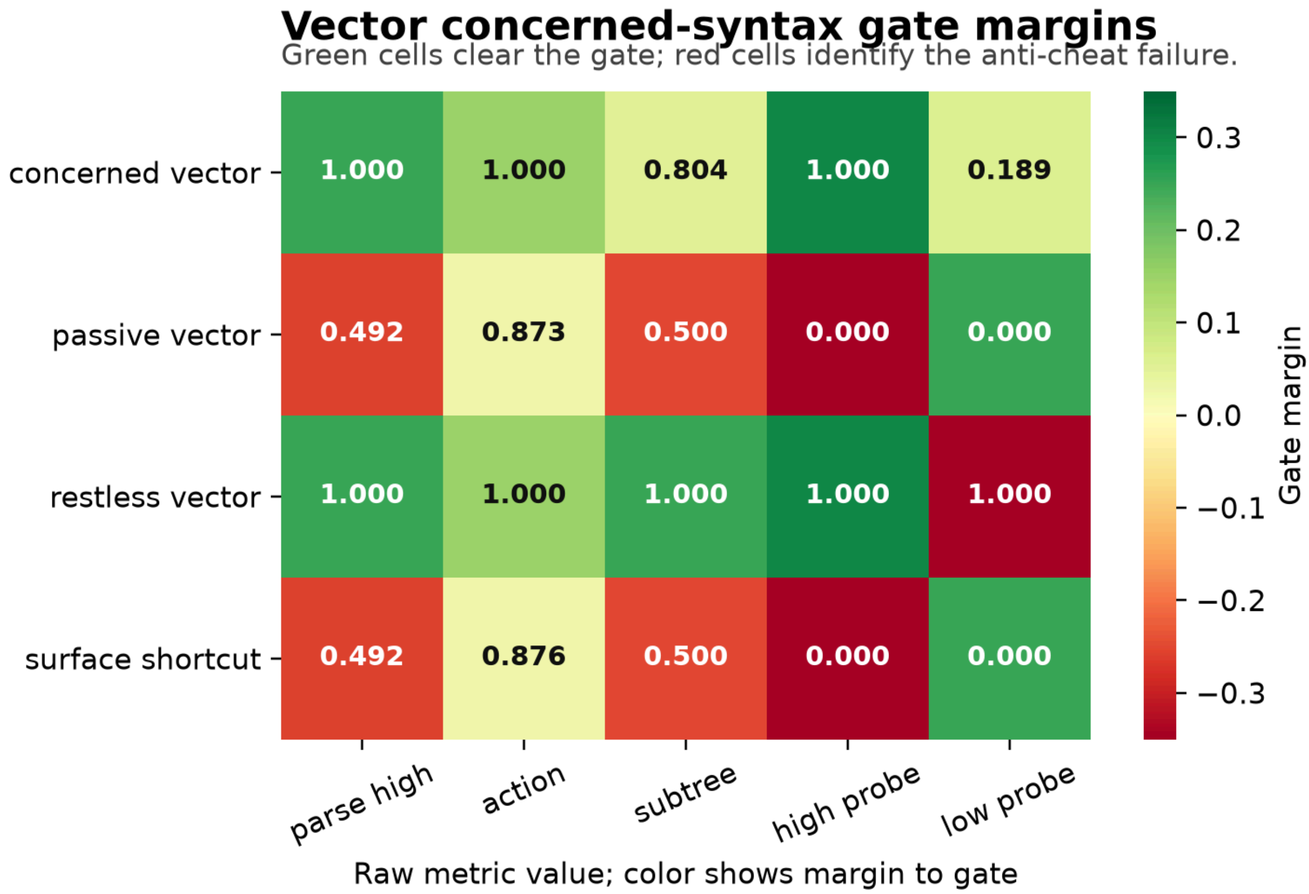
The learned-agent gate still provided candidate parses as visible structural hypotheses. The next gate removes that crutch. Each trial is rendered as a six-part vector surface: role markers, part coordinates, pair identity, and pairwise distances. The surface is deliberately parse-invariant: swapping the hidden true and alternate parse leaves the vector observation unchanged. The only reliable way to recover the causal binding bit is to pay for the pair probe.

Remote command:

```
doppler --scope /Users/jawaun/superoptimizers run -- \
  uvx --python 3.12 --from modal modal run \
  experiments/concerned_syntax/modal_vector_shapes_sweep.py \
  --train-trials 3000 --test-trials 1200 --epochs 90
```

Summary:

Agent	Parse high	Action	Subtree	Low probe	Gate
concerned vector	1.000	1.000	0.804	0.189	PASS
passive vector	0.492	0.873	0.500	0.000	fail
restless vector	1.000	1.000	1.000	1.000	fail
surface shortcut	0.492	0.876	0.500	0.000	fail



This is a stronger anti-cheat surface than the candidate-parse run. Surface and passive vector agents can learn action priors, but they cannot identify which hidden causal binding is active. Restless vector probing recovers syntax while failing concern. The accepted agent passes because it learns when the hidden binding matters and uses the intervention only under a capped calibration guard.

10. Pixel-Rendered Gate

The vector-observation gate still hands the learner explicit generated parts. The next gate renders each six-part surface as a 48x48 RGB image. Role appearance is expressed through color, size, and shape; no candidate parse or hidden binding label is rendered. The extractor computes connected components, then records centroids, areas, mean colors, bounding boxes, densities, and pairwise distances. This is not an unsupervised slot-attention model, but it is a pixel-to-object transition: the learner receives object attributes recovered from pixels, not a symbolic role list or candidate parse.

Local 5-seed command:

```
python3 -m experiments.concerned_syntax.pixel_shapes \
  --train-trials 1200 --test-trials 500 --seed 20260616 --epochs 60 \
  --out artifacts/concerned_syntax/pixel_shapes_local.json \
  --agent-report experiments/concerned_syntax/results/pixel_shapes_local_2026_06_16.md
```

The public local report was produced by the same settings across five seeds: `20260616`, `1729`, `4242`, `8675309`, and `314159`.

Summary:

Agent	Parse high	Action	Subtree	Objects	Low probe	Gate
concerned pixel	0.996	0.999	0.786	1.000	0.187	PASS
passive pixel	0.503	0.874	0.497	1.000	0.000	fail
restless pixel	1.000	0.999	1.000	1.000	1.000	fail
surface pixel shortcut	0.503	0.882	0.497	1.000	0.000	fail

This is the first pixel-level result in Arc 2A. It strengthens the vector claim in one specific way: the surface is now a rendered image, and the perceptual transition is explicit enough to test. It does not strengthen the claim into human-like vision. Passive object extraction still fails because the same image can be generated by multiple hidden parses. Restless probing still fails because syntax without concern probes low-concern ambiguity. The accepted agent passes only when object extraction, concern gating, intervention, and binding are composed.

11. Intervention-Invention Gate

The pixel-rendered gate still provides the target of the intervention: the agent learns when to run a pair probe, but the probe is already aimed at the causal role pair. That is probe use, not intervention invention. The next gate therefore exposes a tiny program menu:

```
null
observe_pair(a,b) for all 15 object pairs
```

The agent does not receive `trial.causal_pair` as metadata. It aligns extracted pixel components to the six visible object positions, scores object slots and candidate pair programs, and composes the selected `observe_pair(a,b)` program with the learned concern gate. This is a deliberately minimal form of intervention invention. It does not yet discover raw motor primitives, but it does force the agent to choose which experiment would make the viability-relevant hidden binding observable.

Remote command:

```
doppler --scope /Users/jawaun/superoptimizers run -- \
  uvx --python 3.12 --from modal modal run \
  experiments/concerned_syntax/modal_intervention_invention_sweep.py \
  --train-trials 3000 --test-trials 1200 --epochs 90
```

The tracked public report is `experiments/concerned_syntax/results/intervention_invention_modal_2026_06_16.md`.

Summary:

Agent	Parse high	Action	Subtree	Low probe	Target high	Useful high	Gate
concerned program inventor	1.000	1.000	0.796	0.156	1.000	1.000	PASS

concern without target	0.534	0.883	0.522	0.156	0.088	0.088	fail
random program probe	0.519	0.879	0.530	1.000	0.060	0.060	fail
surface program shortcut	0.486	0.876	0.494	0.000	0.000	0.000	fail
target without concern	1.000	1.000	1.000	1.000	1.000	1.000	fail

This result sharpens the causal-discovery connection. Recent active causal discovery work treats experimental design as a policy problem and separates prediction from mechanism fidelity. This gate imports the small concern-side version of that demand: the agent must not only predict or probe, but select the intervention target whose observation changes the admissible causal binding under a no-restless constraint.

12. Rich Program-Language Gate

The `observe_pair(a,b)` gate still treats one program family as universally useful. The richer gate makes program-family choice part of intervention invention:

```
null
observe_pair(a,b)
move_anchor(anchor)
ablate_pair(a,b)
compose_move_observe(anchor,a,b)
```

Different role mechanisms require different families. Shield/poison cases require a composed move+observe operation, repair/core cases require move-anchor, food/trap cases require ablation, and low-concern ornament/signal cases should still choose `null`. The accepted agent must therefore compose four decisions: whether concern warrants action, which target matters, which family exposes the mechanism, and how the resulting observation changes action.

Remote command:

```
doppler --scope /Users/jawaun/superoptimizers run -- \
  uvx --python 3.12 --from modal modal run \
  experiments/concerned_syntax/modal_rich_program_language_sweep.py \
  --train-trials 3000 --test-trials 1200 --epochs 90
```

The tracked public report is `experiments/concerned_syntax/results/rich_program_language_modal_2026_06_17.md`.

Gate summary:

Agent	Parse	Action	Subtree	Low program	Gate
concerned composer	1.000	1.000	0.794	0.162	PASS
family no target	0.541	0.890	0.534	0.162	fail
target no family	0.503	0.881	0.633	1.000	fail
rich no concern	1.000	1.000	1.000	1.000	fail
random rich	0.503	0.881	0.507	1.000	fail
surface shortcut	0.503	0.879	0.507	0.000	fail

Program metrics:

Agent	Family	Target	Useful	Rich
concerned composer	1.000	1.000	1.000	1.000
family no target	1.000	0.080	0.080	1.000
target no family	0.000	1.000	0.000	0.000
rich no concern	1.000	1.000	1.000	1.000
random rich	0.249	0.139	0.021	0.749
surface shortcut	0.000	0.000	0.000	0.000

This is stronger than the v1 intervention-invention gate because target selection is no longer enough. The target-only control identifies the right object pair but cannot choose the useful operation family. The rich-without-concern control can choose rich programs but probes low-concern cases restlessly. The accepted agent passes only when concern gating, target binding, program-family selection, and rich program composition are all present.

13. Held-Out Rich Program Transfer Repair

The v2 result above is an in-distribution rich program gate. It proves that the agent can choose the right family and target across seeds, but not that the mechanism survives held-out role-kind and true-parse transfer. The transfer repair gate withholds each high-concern role kind and each true parse family, then evaluates only on the held-out slice.

Remote command:

```
doppler --scope /Users/jawaun/superoptimizers run -- \
  uvx --python 3.12 --from modal modal run \
  experiments/concerned_syntax/modal_rich_program_transfer_repair_sweep.py \
  --train-trials 3000 --test-trials 1200 --epochs 90
```

The tracked public report is `experiments/concerned_syntax/results/rich_program_transfer_repair_modal_2026_06_18.md`.

Gate summary:

Agent	Transfer	Family	Target	Useful	Rich	Low program	Gate
learned rich composer	0.000	0.714	0.829	0.714	0.894	0.161	fail
family only	0.000	1.000	0.196	0.196	1.000	0.000	fail
target only	0.000	0.143	1.000	0.143	0.143	0.714	fail
rich no concern	0.000	1.000	1.000	1.000	1.000	0.714	fail
rich world model	1.000	1.000	1.000	1.000	1.000	0.000	PASS

The learned rich composer remains the shortcut baseline: it can pass the i.i.d. rich-program gate without proving held-out role/parse transfer. The family-only and target-only controls show that neither half of the program decision is sufficient. The rich-without-concern control shows that even the right program family and target become restless without concern gating. The accepted repair decodes role slots, selects the role-equivariant target, chooses the required rich program family, and spends a program only when the decoded world model says the binding matters.

This closes the Modal transfer gate for the provided v2 grammar with an explicit role decoder. The next gate asks whether that decoder can itself be replaced by a learned slot-semantic mapping.

14. Learned Slot-Semantics Transfer

The explicit transfer repair above still knows the RGB role table. The learned slot-semantics gate replaces that nearest-color decoder with a supervised prototype decoder trained from visible role-token calibration examples. The held-out task train/test split is unchanged: each high-concern role kind and each true parse family is withheld from the task learner and evaluated as a transfer slice. The semantic calibration is deliberately narrower than unsupervised object discovery; it learns role-token prototypes from rendered components, then the same world-model operation binds kind, target pair, program family, concern gate, parse observation, and action.

Remote command:

```
doppler --scope /Users/jawaun/superoptimizers run -- \
  uvx --python 3.12 --from modal modal run \
  experiments/concerned_syntax/modal_learned_slot_semantics_sweep.py \
  --train-trials 600 --test-trials 240 --epochs 30 \
  --semantic-calibration-trials 600
```

The tracked public report is `experiments/concerned_syntax/results/learned_slot_semantics_modal_2026_06_18.md`.

Gate summary:

Agent	Transfer	Sem kind	Sem pair	Family	Target	Useful	Rich	Low program
learned slot-semantic world model	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.000
learned rich composer	0.000	0.000	0.000	0.705	0.811	0.705	0.895	0.182
semantic family only	0.000	1.000	1.000	1.000	0.214	0.214	1.000	0.000
semantic target only	0.000	1.000	1.000	0.143	1.000	0.143	0.143	0.714
semantic rich no concern	0.000	1.000	1.000	1.000	1.000	1.000	1.000	0.714

The accepted agent preserves the v2 transfer contract while replacing explicit role decoding with learned supervised slot semantics. The controls preserve the same anti-cheat separations: family without target, target without family, and rich programs without concern all fail despite perfect semantic decoding. This is still not natural-image perception, unsupervised role discovery, or open-ended program invention.

15. Limitations

The current benchmark is still synthetic. The newest agent receives generated pixels with algorithmic connected-component extraction, not natural images, learned object slots, continuous control, or human subjects. The newest rich-program gate selects among a provided finite program grammar; it does not invent raw motor primitives or open-ended experimental apparatus. The point of this first paper is to define and pass a minimal learned acceptance surface before larger compute.

The Modal transfer and learned slot-semantics repairs narrow the next limitation but do not erase it: the agent should learn object slots without supervised role-token calibration, invent or search programs beyond the provided grammar, and instantiate program/body components as learned modules rather than compact hand-instantiated world-model operations.

16. Conclusion

Arc 2A inserts a new layer into the maintained-concern ladder:

```
difference -> geometry -> syntax -> salience -> valence
          -> action -> attribution -> maintenance
```

The results support a narrow methodological claim: concerned syntax needs its own tests. Reward, compression, uncertainty, action accuracy, target selection without concern, rich program use without concern, and even syntax without concern can all dissociate from causal constituency under maintained concern. The learned-agent, vector-observation, pixel-rendered, intervention-invention, and rich-program gates show that the gate is passable without hidden parse access, but only when object extraction, useful target selection, program family selection, binding, intervention, and formal concern gating are present together. The held-out v2 transfer and learned slot-semantics repairs add the next boundary: the same grammar can survive role/parse transfer when a world-model concern gate consumes either explicit or supervised learned role semantics. Unsupervised object slots, program discovery beyond the provided grammar, and learned module/body search remain future work.

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